

D - Card Pile Query

分值: 400400400 分

Problem Statement

There are NNN cards and NNN piles of cards.

有 NNN 张卡片和 NNN 堆卡片。

The cards and the piles are numbered $1, 2, \dots, N, 1, 2, \dots, N, 1, 2, \dots, N$. Initially, pile iii contains only card iii .

卡片和堆的编号为 $1, 2, \dots, N, 1, 2, \dots, N, 1, 2, \dots, N$ 。初始时, 第 iii 堆只有卡片 iii 。

Perform the following operation for each $i = 1, 2, \dots, Q, i = 1, 2, \dots, Q, i = 1, 2, \dots, Q$ in order:

按顺序对每个 $i = 1, 2, \dots, Q, i = 1, 2, \dots, Q, i = 1, 2, \dots, Q$ 执行如下操作:

- Move card $C_i C_i C_i$ and all cards stacked on top of card $C_i C_i C_i$, preserving their order, on top of card $P_i P_i P_i$. It is guaranteed that immediately before performing the operation, cards $C_i C_i C_i$ and $P_i P_i P_i$ are in different piles, and card $P_i P_i P_i$ is on top of some pile.
- 将卡片 $C_i C_i C_i$ 以及叠在它上面的所有卡片保持原有顺序, 移到卡片 $P_i P_i P_i$ 的顶部。题目保证执行该操作前, 卡片 $C_i C_i C_i$ 和 $P_i P_i P_i$ 位于不同的堆中, 且卡片 $P_i P_i P_i$ 位于某一堆的顶部。

Find the number of cards in each pile after all operations are completed.

求所有操作完成后每一堆的卡片数量。

Constraints

- $1 \leq N, Q \leq 3 \times 10^5, 1 \leq N, Q \leq 3 \times 10^5, 1 \leq N, Q \leq 3 \times 10^5$
- $1 \leq C_i, P_i \leq N, 1 \leq C_i, P_i \leq N, 1 \leq C_i, P_i \leq N$
- When the operations are performed in order, immediately before each operation, cards $C_i C_i C_i$ and $P_i P_i P_i$ are in different piles.
- When the operations are performed in order, immediately before each operation, card $P_i P_i P_i$ is on top of some pile.

- All input values are integers.
- $1 \leq N, Q \leq 3 \times 10^5$
- $1 \leq C_i, P_i \leq N$
- 按顺序执行操作时，每次操作前，卡片 $C_i C_i C_i$ 和 $P_i P_i P_i$ 位于不同的堆中。
- 按顺序执行操作时，每次操作前，卡片 $P_i P_i P_i$ 位于某一堆的顶部。
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按如下格式从标准输入给出：

```

NNN QQQ
C1C1C1 P1P1P1
C2C2C2 P2P2P2
⋮
CQCQ PQPQ

```

Output

Let A_i be the number of cards in pile i at the end. Output

$A_1, A_2, \dots, A_N$ in this order, separated by spaces.

设 A_i 为最终第 i 堆的卡片数量。按顺序输出

$A_1, A_2, \dots, A_N$ ，数值之间用空格分隔。

Sample Input 1

```

5 4
1 3
4 5
1 4
4 2

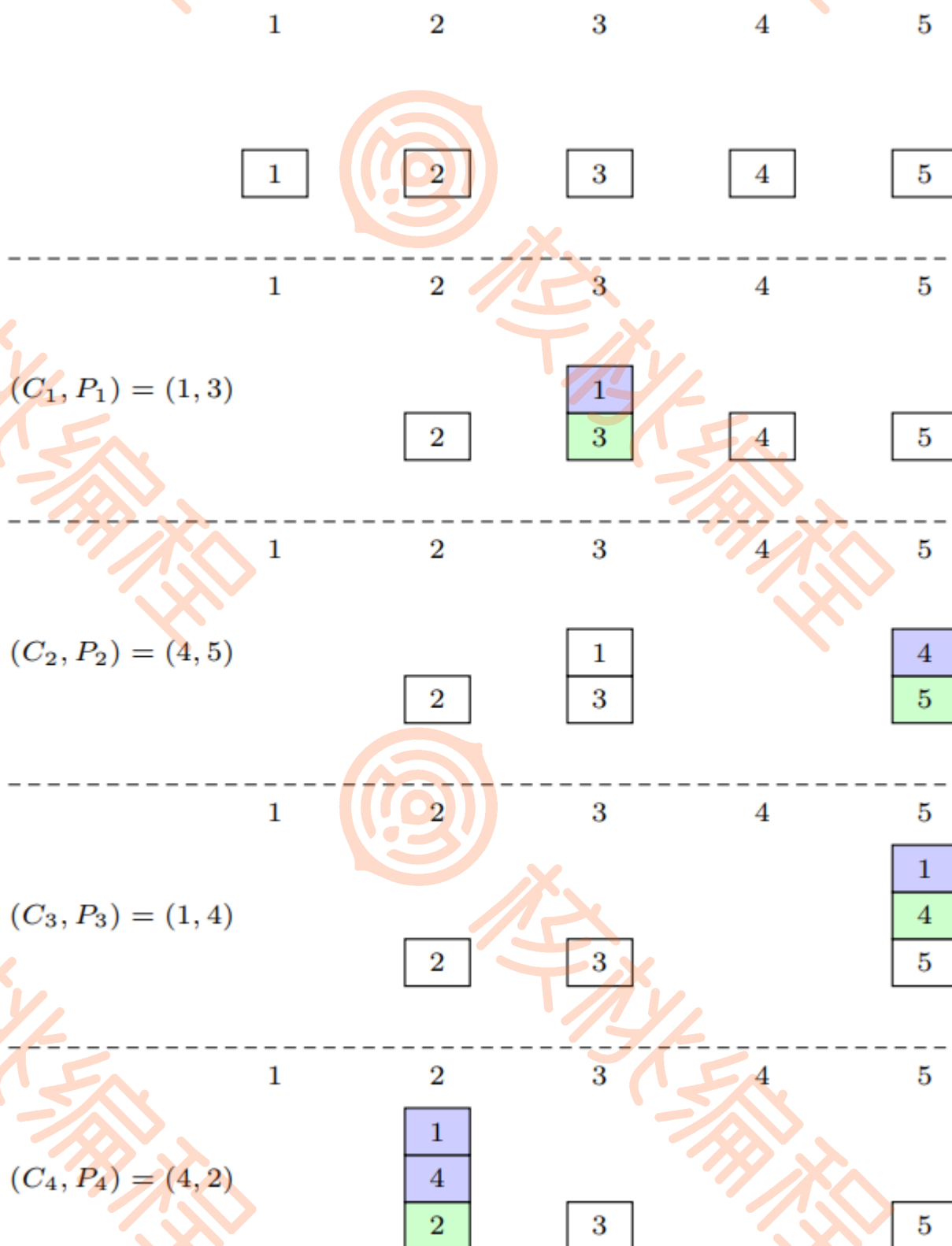
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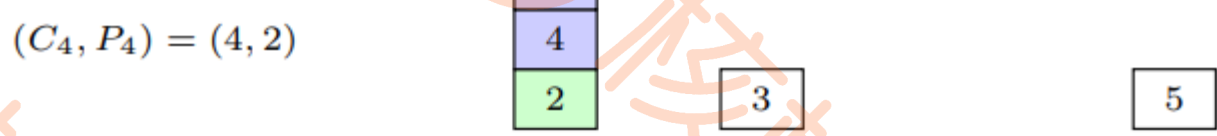
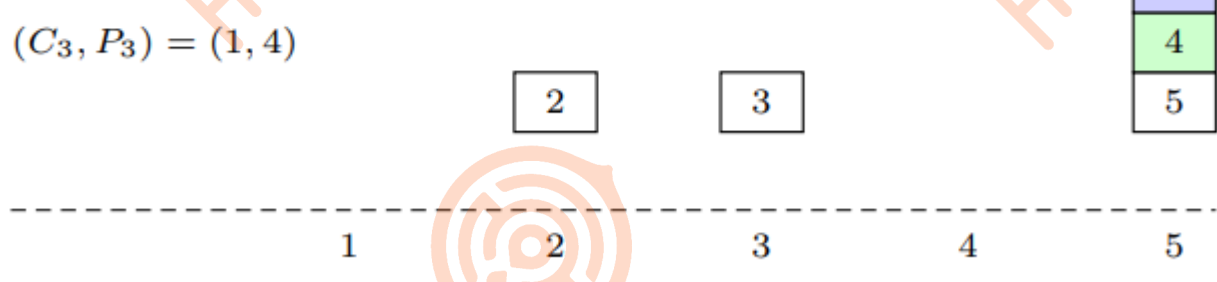
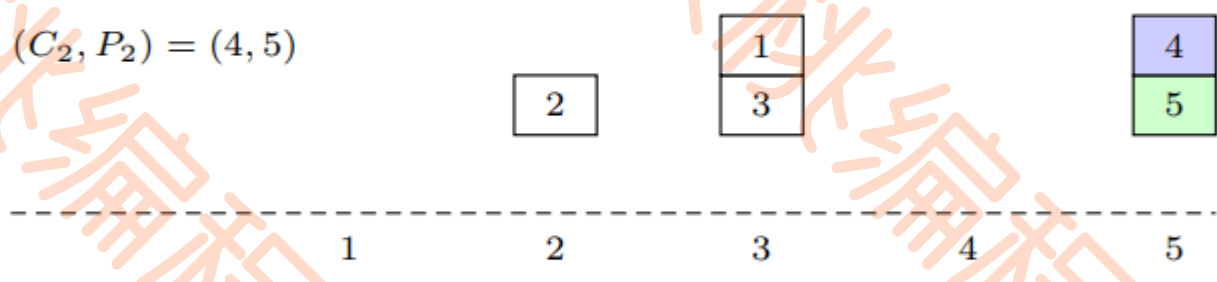
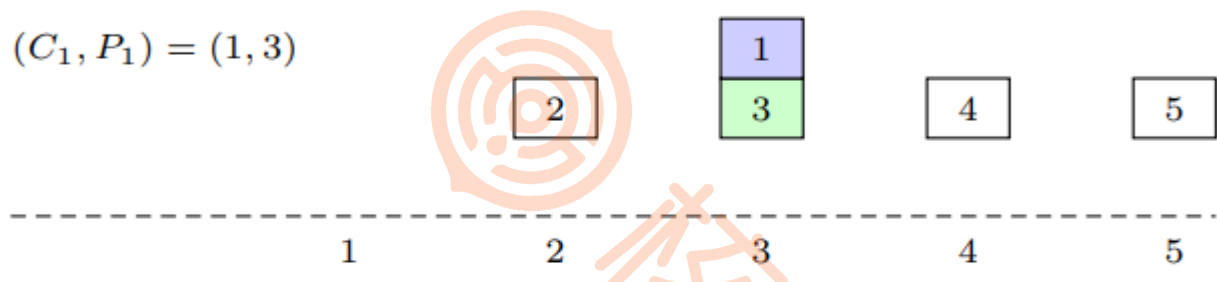
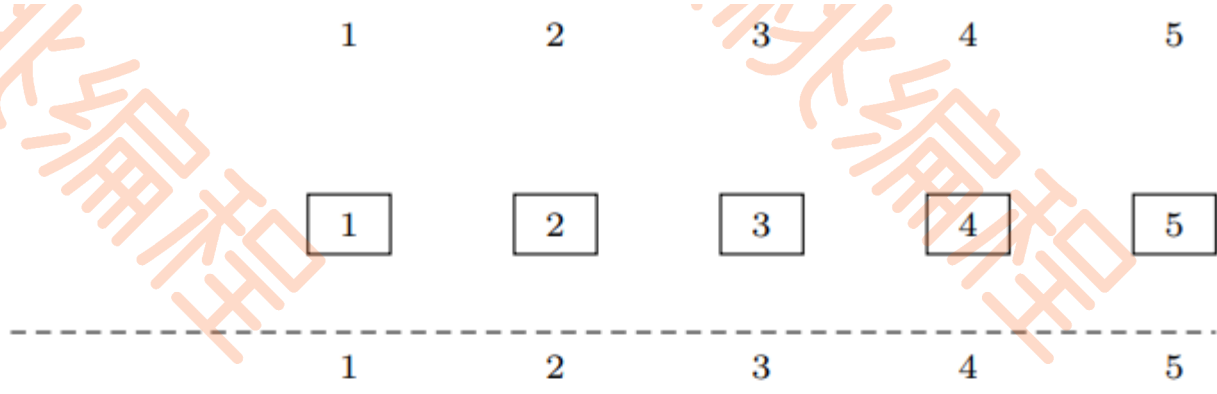
Sample Output 1

0 3 1 0 1

The sequence of operations proceeds as shown in the figure below.

操作序列的执行过程如下图所示。





Sample Input 2

```
7 8
3 1
5 4
```

```
2 5
5 7
2 3
6 2
3 4
5 1
```

Sample Output 2

```
2 0 0 4 0 0 1
```